

Matgeo Presentation - Problem 12.602

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Question

If the following system has a non-trivial solution:

$$px + qy + rz = 0,$$

$$qx + ry + pz = 0,$$

$$rx + py + qz = 0,$$

then which one of the following options is **TRUE**?

(CS 2016)

- (a) $p + q + r = 0$ or $p = q = r$
- (b) $p + q + r = 0$ or $p = q = -r$
- (c) $p + q + r = 0$ or $p = q = r = 0$
- (d) $p + q + r = 0$ or $p = -q = r$

Solution

Given equations are

$$(p \quad q \quad r) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \quad (0.1)$$

$$(q \quad r \quad p) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \quad (0.2)$$

$$(r \quad p \quad q) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \quad (0.3)$$

$$\Rightarrow \begin{pmatrix} p & q & r \\ q & r & p \\ r & p & q \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad (0.4)$$

Solution

let

$$\mathbf{A} = \begin{pmatrix} p & q & r \\ q & r & p \\ r & p & q \end{pmatrix} \text{ and } \mathbf{X} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad (0.5)$$

$$\implies \mathbf{AX} = \mathbf{0} \quad (0.6)$$

The system of equations $\mathbf{AX} = \mathbf{0}$ has non trivial solution if rank of $\mathbf{A} < 3$

$$\begin{pmatrix} p & q & r \\ q & r & p \\ r & p & q \end{pmatrix} \xleftrightarrow{R_2 \rightarrow pR_2 - qR_1} \begin{pmatrix} p & q & r \\ 0 & pr - q^2 & p^2 - qr \\ r & p & q \end{pmatrix} \quad (0.7)$$

$$\xleftrightarrow{R_3 \rightarrow pR_3 - rR_1} \begin{pmatrix} p & q & r \\ 0 & pr - q^2 & p^2 - qr \\ 0 & p^2 - rq & pq - r^2 \end{pmatrix} \quad (0.8)$$

$$\xleftrightarrow{R_3 \rightarrow (pr - q^2)R_3 - (p^2 - rq)R_2} \begin{pmatrix} p & q & r \\ 0 & pr - q^2 & p^2 - qr \\ 0 & 0 & D \end{pmatrix} \quad (0.9)$$

Solution

$$\implies D = (pr - q^2)(pq - r^2) - (p^2 - rq)(p^2 - qr) = 0 \quad (0.10)$$

$$\implies (pr - q^2)(pq - r^2) = p^2qr - pr^3 - pq^3 + q^2r^2 \quad (0.11)$$

$$\implies (p^2 - rq)(p^2 - qr) = p^4 - 2p^2qr + r^2q^2 \quad (0.12)$$

From (0.11) and (0.12)

$$D = (p^2qr - pr^3 - pq^3 + q^2r^2) - (p^4 - 2p^2qr + r^2q^2) \quad (0.13)$$

$$= -p^4 + 3p^2qr - pq^3 - pr^3 \quad (0.14)$$

$$= -p(p^3 + q^3 + r^3 - 3pqr) \quad (0.15)$$

$$= -p(p + q + r)(p^2 + q^2 + r^2 - pq - qr - pr) \quad (0.16)$$

$$\implies D = -\frac{1}{2}p(p + q + r) \left((p - q)^2 + (q - r)^2 + (p - r)^2 \right) \quad (0.17)$$

$D = 0$ means

$$p + q + r = 0 \quad \text{or} \quad (p - q)^2 + (q - r)^2 + (p - r)^2 = 0 \quad (0.18)$$

$$\implies p + q + r = 0 \quad \text{or} \quad p = q = r \quad (0.19)$$

Conclusion

$\Rightarrow$ Given system of equations has non trivial solution if

$$p + q + r = 0 \quad \text{or} \quad p = q = r \quad (0.20)$$

C Code: program.c

```
#include <stdio.h>
#include <math.h>

int main() {
    FILE *fp;

    // Open output file
    fp = fopen("answer.dat", "w");
    if (fp == NULL) {
        printf("Error opening file!\n");
        return 1;
    }

    // ----- Example 1:  $p + q + r = 0$  -----
    double p1 = 1, q1 = 2, r1 = -3; // satisfies  $p + q + r = 0$ 
    fprintf(fp, "Example 1:\n");
    fprintf(fp, "Given values: p=%%.2lf, q=%%.2lf, r=%%.2lf\n", p1, q1, r1);

    if (fabs(p1 + q1 + r1) < 1e-9 || (fabs(p1 - q1) < 1e-9 && fabs(q1 - r1) < 1e-9)) {
        fprintf(fp, "The system has a NON-TRIVIAL solution.\n");
        if (fabs(p1 + q1 + r1) < 1e-9)
            fprintf(fp, "Condition satisfied: p+q+r=0\n\n");
        else
            fprintf(fp, "Condition satisfied: p=q=r\n\n");
    } else {
        fprintf(fp, "The system does NOT have a non-trivial solution.\n\n");
    }
}
```

C Code: program.c

```
// ----- Example 2:  $p = q = r$  -----
double p2 = 5, q2 = 5, r2 = 5; // satisfies  $p = q = r$ 
fprintf(fp, "Example2:\n");
fprintf(fp, "Given values: p=%0.2lf, q=%0.2lf, r=%0.2lf\n", p2, q2, r2);

if (fabs(p2 + q2 + r2) < 1e-9 || (fabs(p2 - q2) < 1e-9 && fabs(q2 - r2) < 1e-9)) {
    fprintf(fp, "The system has a NON-TRIVIAL solution.\n");
    if (fabs(p2 + q2 + r2) < 1e-9)
        fprintf(fp, "Condition satisfied: p+q+r=0\n");
    else
        fprintf(fp, "Condition satisfied: p=q=r\n");
} else {
    fprintf(fp, "The system does NOT have a non-trivial solution.\n");
}

fclose(fp);
printf("Results for both examples written to answer.dat\n");
return 0;
}
```


Python: plot.py

```
import numpy as np

# Example 1: satisfies  $p + q + r = 0$ 
p, q, r = 1, 2, -3

# Example 2 (uncomment to test  $p=q=r$  case)
# p, q, r = 5, 5, 5

# Construct coefficient matrix A
A = np.array([
    [p, q, r],
    [q, r, p],
    [r, p, q]
], dtype=float)

# Compute determinant
detA = np.linalg.det(A)

print(f"Given: p={p}, q={q}, r={r}")
print("Matrix A=\n", A)
print(f"Determinant of A={detA:.6f}")

# Check for non-trivial solution
if abs(detA) < 1e-9:
    print("\nThe system has a NON-TRIVIAL solution.")
    if abs(p + q + r) < 1e-9:
        print("Condition satisfied:  $p+q+r=0$ ")
    elif abs(p - q) < 1e-9 and abs(q - r) < 1e-9:
        print("Condition satisfied:  $p=q=r$ ")
    else:
        print("Condition satisfied: determinant=0 (degenerate case)")
else:
    print("\nThe system has only the TRIVIAL solution ( $x=y=z=0$ ).")
```